

Ligjerata 3d

Hedhja vertikale

$$v_0 \neq 0$$

$$\vec{v} = \vec{v}_0 - \vec{v}_g$$

$$v_g = gt$$

$$v = v_0 - gt$$

$$v = \frac{dh}{dt}$$

$$\frac{dh}{dt} = v_0 - gt$$

$$dh = (v_0 - gt) dt$$

$$dh = v_0 dt - gt dt \quad \int$$

$$h = v_0 t - \frac{gt^2}{2}$$

Kese $h = H \quad v = 0$

$$v = v_0 - gt$$

$$0 = v_0 - gt \Rightarrow v_0 = gt \Rightarrow t = \frac{v_0}{g}$$

$$h = v_0 t - \frac{gt^2}{2}$$

$$H = v_0 \frac{v_0}{g} - g \frac{v_0^2}{2g^2} = \frac{v_0^2}{g} - \frac{v_0^2}{2g}$$

$$H = \frac{v_0^2}{2g}$$

